

Evidence-Gated Medical LLM Alerts for Pharmacogenomic Claims

Anonymous Author(s)

Abstract

LLM-generated medication alerts can sound more certain than their evidence permits. We built a deterministic pre-display monitor for one workflow: pharmacogenomic and genomic medication-alert text. The monitor consumes structured annotations for citation/guideline support, population fit, and endpoint/actionability, then returns allow, narrow, abstain, or deny. Stage 1 does not call an LLM or parse free text; it tests whether the three-part decision boundary is internally necessary before live extraction is attempted. Disabling citation/guideline support let 2 of 23 designed overclaims pass unchanged; disabling population fit let 1 pass; disabling endpoint/actionability let 6 pass. With all gates active, all 23 designed overclaim archetypes were blocked or narrowed, and all 10 bounded alerts passed unchanged. The first-triggering gate matched the author-assigned gate label in 31 of 33 cases. These are specification-conformance results, not clinical safety, generalization, or patient-benefit evidence.

Keywords: clinical decision support; large language models; pharmacogenomics; evidence overclaiming; model evaluation

Data and Code Availability The Stage 1 case bank is synthetic and contains no real patient records, protected health information, or private genomic data. For anonymous venue review, code and synthetic data should be supplied as an anonymized supplemental archive; the de-anonymized public repository may be disclosed after acceptance or in course-facing materials.

Institutional Review Board (IRB) Stage 1 uses only author-designed synthetic cases and public guideline/database names, and therefore does not require IRB approval. Any Stage 2 use

of clinical cases or expert adjudication would require appropriate local governance.

1. Introduction

Healthcare LLMs can make evidentiary overreach look syntactically clean. A medication alert may cite a real guideline while turning a bounded pharmacogenomic recommendation into a universal instruction, a population-specific association into a claim for a different population, or a surrogate endpoint into patient-outcome benefit. Here, transportability means applying evidence from one source population as if it supported the same claim in a different target population. The gap matters because fluency can hide what evidence does, and does not, authorize. Pharmacogenomic medication alerts are a useful testbed because CPIC and ClinPGx already distinguish genotype, phenotype, evidence level, and prescribing recommendation ([Clinical Pharmacogenetics Implementation Consortium, 2026](#); [ClinPGx, 2026a](#)).

2. Evidence Boundary

Healthcare AI evaluation should distinguish implementation consistency from patient-outcome benefit. Byrne et al. emphasize that outcome claims require pragmatic clinical evaluation ([Byrne et al., 2024](#)), while Stegenga motivates caution when evidentiary pipelines make interventions appear stronger than they are ([Stegenga, 2018](#)). FDA–NIH BEST materials provide endpoint vocabulary: biomarkers, surrogate endpoints, validated surrogates, and clinical outcomes are not interchangeable ([U.S. Food and Drug Administration and National Institutes of Health, 2016](#)). Pharmacogenomics adds a positive control: CPIC-style resources can support bounded prescribing

language, including CYP2C19–clopidogrel and DPYD–fluoropyrimidine examples (Lee et al., 2022; Amstutz et al., 2018), but do not automatically support claims that an LLM improves outcomes, that an association is causal, or that a result generalizes across populations without recalibration.

3. Related Work

LLM factuality methods such as SelfCheckGPT test whether generated content is consistent across black-box samples (Manakul et al., 2023). Selective classification supplies the decision vocabulary for abstention: a system may decline to answer when coverage is less important than risk control (Geifman and El-Yaniv, 2017). Clinical alert-fatigue work explains why this choice matters in practice: repeated and low-information alerts can reduce clinician response (Ancker et al., 2017). This study tests a smaller executable claim. Given structured annotations about source support, target population, and endpoint strength, a pre-display monitor can prevent designed pharmacogenomic overclaims without suppressing bounded alerts.

4. Methods

The controller is a deterministic pre-display monitor, not an autonomous clinical agent. It receives a synthetic candidate alert plus structured evidence annotations assigned in the case bank; it does not parse draft text, call an LLM, verify citations, or extract EHR fields or population descriptors. The fixed precedence order is citation/guideline support, population fit, then endpoint/actionability. It returns one of five actions: allow, narrow, abstain (population fit), abstain (citation support), or deny.

5. Stage 1 Scaffold

The case bank contains 33 author-designed synthetic cases: 10 bounded aligned alerts and 23 overclaim archetypes involving universal action language, unsupported genetics, population mismatch, citation weakness, variant uncertainty, obsolete evidence, and changed clinical context. An isolating case is designed so that one gate is

responsible for blocking the overclaim while the other two gates would not independently block it. Within the 23 overclaim archetypes, three cases isolate a fabricated CPIC addendum attached to an otherwise actionable claim, a real guideline-supported claim applied to a mismatched population, and a stale but real guideline used as current evidence. The 23/10 split is stress-test coverage, not a clinical base rate. Each case has expected labels, but the controller does not read them. The three checks are summarized in Table 1.

All 23 designed overclaim archetypes were blocked or narrowed and all 10 bounded alerts passed unchanged, so the controller is not a blanket denial rule. Table 2 shows that disabling each gate allows at least one designed overclaim to pass unchanged, so the gates are not merely duplicating endpoint/actionability. Removing citation/guideline support allows 2/23 overclaims through unchanged; removing population fit allows 1/23 through; removing endpoint/actionability allows 6/23 through. No gate is redundant in this synthetic case bank: each gate blocks at least one designed overclaim that passes unchanged when that gate is disabled.

Primary-gate conformance means that the first enabled gate named by the controller matches the author-assigned gate label. It was 31/33. In PGX19 and PGX24, the controller returned the expected action, but the triggering gate differed from the author-assigned gate label because citation/guideline support has precedence. These two disagreements are gate-attribution errors.

6. Evaluation Plan

Stage 2 is a held-out comparison, not a patient-benefit claim. Independent case authors write pharmacogenomic alert scenarios without seeing the decision table. The ungated arm shows the LLM draft unchanged; the gated arm shows the same draft after structured annotations pass through the monitor. Two clinical reviewers score overclaiming, inappropriate denial, preserved clinician authority, population fit, citation support, and error category. The primary analysis estimates overclaim reduction with a prespecified upper bound on inappropriate denial.

Table 1: Stage 1 decision predicates implemented by the controller.

Gate	Predicate used in Stage 1	Action
citation/guideline support	Source cannot be verified.	Deny citation.
citation/guideline support	Source is conflicting, uncertain, outdated, or applied after the clinical context has changed.	Abstain (citation support).
citation/guideline support	Source support is absent. If the requested medication action is also not supported, deny; otherwise abstain.	Deny unsupported action or abstain (citation support).
population fit	Target-population fit is mismatched or cannot be determined after citation/guideline support passes.	Abstain (population fit).
endpoint/actionability	The requested action is not supported for the claim.	Deny unsupported action.
endpoint/actionability	The claim depends on context, association evidence, pharmacology alone, a dose algorithm, a surrogate endpoint, or an uncertain variant.	Narrow.
none	Earlier predicates do not fire.	Allow bounded alert.

Table 2: One-gate ablation on the synthetic case bank.

Disabled gate	Allowed	Action Δ	Gate Δ
Citation/guideline support	2	7	12
Population fit	1	5	5
Endpoint/actionability	6	6	6

ment recommendation; deployed clinical decision support would still need accountability for extraction, citation verification, thresholds, overrides, and audit logging. The key limitation is that results are synthetic and author-designed. They show rule conformance, not generalization, and assume structured annotations rather than extracting them from live LLM output or EHR context.

7. Ethics and Limitations

Transparency The gate produces a decision trace. For each alert, the trace names the structured evidence feature that triggered allow, narrow, abstain, or deny. That trace is the governance object: clinicians and reviewers can inspect the rule that changed the displayed alert.

Fairness Fairness and bias are central in pharmacogenomics. HLA-B*15:02 risk language depends on ancestry and tested population, founder-variant evidence may not generalize to the US population, and biobank results may be unrepresentative for underrepresented or recently admixed groups (ClinPGx, 2026b; Genome India Project, 2026). Cases PGX09, PGX10, PGX25, PGX26, and PGX27 treat population fit as an equity constraint. PGX24 represents the high-risk practice of using a polygenic score to deny opioid or stimulant therapy.

Governance and limitations Stage 1 contains no PHI and makes no diagnosis or treat-

8. Conclusion

The practical problem is annotation, not compute. If evidence support, target population, and endpoint strength can be extracted reliably from live LLM output and EHR context, a deterministic pre-display monitor gives reviewers an auditable place to stop overclaiming before it reaches the clinician. On the synthetic Stage 1 case bank, citation/guideline support, population fit, and endpoint/actionability each block at least one designed overclaim that passes when that gate is removed.

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